$A_1$ = Selection of black socks,

$$n(A_1) = 3C_2 = 3$$

$$P(A_1) = \frac{n(A_1)}{n(S)} = \frac{3}{21}$$

$A_2$ = Selection of blue socks,

$$n(A_2) = 4C_2 = 6$$

$$P(A_2) = \frac{n(A2)}{n(S)} = \frac{6}{21}$$

then $A_1 \cup A_2$ represents the event of selecting 2 socks of same colour. Since the occurrence of one event excludes the occurrence of the other, these two events are mutually exclusive. Then, by Axiom-3,

$P(A_1 \cup A_2) = P(A_1) + P(A_2)$

Therefore, $P(A_1 \cup A_2) = \frac{3}{21} + \frac{6}{21} = \frac{9}{21} = \frac{3}{7}$

Thus, the probability of selecting two socks of same colour is 3/7.

**Example 8.11**

Angel selects three cards at random from a pack of 52 cards. Find the probability of drawing:

(i) 3 spade cards.

(ii) one spade and two knave cards

(iii) one spade, one knave and one heart cards.

***Solution:***

Total no. of ways of drawing 3 cards = n(S) = 52 $C_3$ = 22100

(a) Let $A_1$ = drawing 3 spade cards.

Since there are 13 Spades cards in a pack of cards,

No. of ways of drawing 3 spade cards = n($A_1$) = 13 $C_3$ = 286

Therefore, $P(A_1) = \frac{n(A_1)}{n(S)} = \frac{286}{22100}$

(b) Let $A_2$ = drawing one spade and two knave cards

No. of ways of drawing one spade card = $13C_1$ = 13

No. of ways of drawing two knave cards = $13C_2$ = 78